22 PROGRAMMING PRACTICE • THEORY

01 mathematical computing problem solving

Detailed Theory, Exact C++ Rules, Memory Models & Exam Reference

01 — Mathematical Computing Problem-Solving in C++23

Component-Wise Addition of Quadratic Polynomials

```
p1 = 3x^2 + 4x + 5 | p2 = 2x^2 + 6x + 1

p1 + p2 = (3+2)x^2 + (4+6)x + (5+1) = 5x^2 + 10x + 6
```

Polynomial Addition

In B.Tech Mathematics and Computing examinations, classic mathematical structures — **Quadratic Polynomials**, **Rational Fractions (p/q)**, **Complex Numbers ($a+bi$)**, and **Matrices** — are modeled as simple classes with constructors, `operator+`, `operator*`, and `display()` member functions.